

Working Example

Example 9.8 Trace metals in drinking water affect the flavor, and unusually high concentrations can pose a health hazard. The article “Trace Metals of South Indian River” (*Envir. Studies*, 1982: 62–66) reports on a study in which six river locations were selected (six experimental objects) and the zinc concentration (mg/L) determined for both surface water and bottom water at each location. The six pairs of observations are displayed in the accompanying table. Does the data suggest that true average concentration in bottom water exceeds that of surface water?

	Location					
	1	2	3	4	5	6
Zinc concentration in bottom water (x)	.430	.266	.567	.531	.707	.716
Zinc concentration in surface water (y)	.415	.238	.390	.410	.605	.609
Difference	.015	.028	.177	.121	.102	.107

Devore [2019]

Jay L. Devore. *Probability and Statistics for Engineering and the Sciences*. 8. ed edition, 2019.